	Horizon Campus **Faculty of Information Technology** **BSc (Hons.) in Information Technology/BIT (Hons.) in Networking and Mobile Computing**

Please fully complete all sections as requested.

Degree Program: BIT (HONS)NETWORKING AND MOBILE COMPUTING BSC(HONS)IN INFORMATION TECHNOLOGY		
Research Details		
Student Details / Group Name:		
Name with Initials	Registration No	Phone No
A.M.K.D ADHIKARI	ITBIN-2211-0137	071-3256156
W.THILAN AVISHKA	ITBNM-2211-0199	075-5693410
Primary Supervisor		
Name with Initials	Email	Phone No
Ms A.A.P.W. Athukorala	parami@horizoncampus.edu.lk	0781607826
Co-supervisor / Mentor (if any)		
Name with Initials	Email	Phone No
Mr.Daminda Herath	damindaherath@horizoncampus.edu.lk	0777296275
Coordinating supervisor		
Name with Initials	Email	Phone No
Proposed Research Project Details:		
1. **Working Title**		
SmartWaste Pro – A Sensor-Based Intelligent Waste Marketplace and Monitoring System		

2. Problem Statement

In many areas across Sri Lanka, waste is often disposed of in a disorganized way, without separation or tracking. While automatic sorting of waste using robotics or AI is possible in advanced systems, it is not practical in distributed rural or institutional environments due to cost and complexity. On the other hand, manual separation using separate bin compartments is more realistic and widely accepted. However, most bins do not offer real-time fill-level monitoring or waste value tracking. This project introduces a smart bin with three manual compartments (e.g., plastic, food, metal) and sensors to monitor each compartment's status. It connects to a mobile platform that allows registered buyers and suppliers to interact, track waste availability, and manage pickups based on waste types and buyer preferences.

3. Research Aim

To develop a sensor-powered smart bin system with multiple waste compartments and a connected marketplace app that automates monitoring, notification, and buyer-supplier interactions for recyclable materials.

4. Research Objectives

Objective 1: To design a three-compartment waste bin with sensors in each section to monitor fill level and weight individually.

Objective 2: To develop a mobile/web platform that enables waste suppliers and buyers to register, receive alerts, and track waste availability.

Objective 3: To implement efficient communication protocols for waste data updates using energy-aware IoT techniques such as EGANT.

5. Research Questions

Question 1: How can we detect and transmit waste levels separately in a multi-compartment bin setup without using complex image-based AI?

Question 2: What platform features can automate fair buyer selection, price visibility, and pickup scheduling in a decentralized waste marketplace?

Question 3: How can we minimize power consumption in a network of bins using energy-efficient data transmission and logic processing?

6. Background Literature Review

Waste management is a growing challenge in urban and semi-urban areas of developing countries like Sri Lanka. Traditional bins lack the ability to report their status, making waste collection inefficient and unscheduled. While automated waste sorting using computer vision and robotics exists in some parts of the world, such systems are expensive, maintenance-heavy, and impractical for wide deployment. Research supports a simpler and more realistic approach: manual sorting combined with sensor-based monitoring.

According to a study published in *IEEE Xplore* (2022), smart bins that use **IR and ultrasonic sensors** have been successfully implemented in cities like Seoul and Singapore to monitor bin fill levels. These systems alert waste collectors only when bins are full, significantly reducing unnecessary collection trips and fuel usage. Such techniques can be adapted to work in bins with **separate compartments**, allowing each waste type (e.g., food, plastic, metal) to be monitored independently.

In a study by Y. Zhao et al. (*SpringerLink*, 2021), researchers showed that **moisture sensors and weight sensors** are useful for identifying organic waste (like food) based on its high water content and lower weight compared to metal or plastic. These sensors can be deployed in the food waste compartment to monitor when it needs to be emptied, helping avoid odor and microbial build-up.

Research from *ScienceDirect* (2020) highlighted the role of **inductive sensors** in detecting metallic objects. When users place metal waste in the assigned compartment, these sensors can confirm the presence of metal without needing camera-based identification. This method is energy-efficient and cost-effective, especially when integrated with basic microcontrollers like the ESP32.

From a communication perspective, modern smart bin systems must balance performance with energy consumption. A recent paper on **EGANT (Energy-efficient Graph Attention Network)** from *IEEE Xplore* (2023) describes how energy-aware routing in wireless sensor networks reduces transmission frequency without losing important data. This technique is especially useful in systems with many distributed bins where power consumption is a concern.

Additionally, platforms that enable **buyers and sellers of recyclable waste** have been studied in publications from the *ACM Digital Library*. These systems allow small-scale recyclers to register, receive alerts, and offer prices for materials like plastic or paper. The findings show that integrating a **marketplace model** with waste monitoring tools increases participation and supports the local circular economy.

In summary, recent research shows that using **sensor-based detection in fixed waste compartments** is a feasible and scalable method for waste categorization and monitoring. Combined with IoT-based communication and a digital marketplace for buyers, this model provides both environmental and economic benefits, especially for developing regions.

7. Data Requirement

Primary Data:

- Sensor data from each compartment (weight, level, moisture)
- App activity (logins, waste offers, bids, pickups)

Secondary Data:

- Waste material pricing trends
- IoT transmission methods and power optimization research
- Public waste bin design case studies

8. Hardware/ Software Requirements

Hardware:

- ESP32 microcontroller
- IR sensors (3) – one for each compartment
- Load cell modules (optional – for accuracy)
- Rain/moisture sensor – for organic bin
- Buzzer + LED
- Wi-Fi / LoRa modules

All essential components will be acquired during the **first 10–12 weeks** of the project using personal funds. If needed, I will use university lab access or consult with the supervisor for for testing on campus.

Software:

- Arduino IDE / PlatformIO
- Firebase or MQTT server
- React Native / Flutter (app)
- Firestore (for real-time database)
- Node-RED / n8n for automation and workflows

All software tools used are **open-source or free-tier cloud platforms**. Firebase and Firestore offer generous free plans suitable for student projects. I will start the software phase in **month 2 and 3**, alongside the hardware build.

9. Ethics

This project uses only general sensor data and does not involve people, private information, or animals. It encourages environmental sustainability and supports ethical recycling through a fair system.

10. References

[1] M. Alenezi, M. Alshammari and N. Alabduljabbar, “Smart waste management system using IoT technology,” *IEEE Access*, vol. 9, pp. 14026–14040, 2021. [Online]. Available: https://ieeexplore.ieee.org/document/9355035

[2] Y. Zhao, J. Liu, and H. Ma, “Moisture-based classification of organic waste using low-cost sensors,” *Environmental Monitoring and Assessment*, vol. 193, no. 4, pp. 1–10, 2021. [Online]. Available: https://link.springer.com/article/10.1007/s10661-021-08989-4

[3] D. Singh, A. Jain, and K. Sharma, “Energy-efficient communication protocol for smart waste bins using EGANT,” *IEEE Sensors Journal*, vol. 23, no. 2, pp. 1431–1440, Jan. 2023. [Online]. Available: https://ieeexplore.ieee.org/document/9987651

[4] R. Gutiérrez, L. Gómez, and M. Perez, “Application of inductive sensors for smart material separation in waste bins,” *Waste Management*, vol. 107, pp. 76–83, 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0956053X2030050X

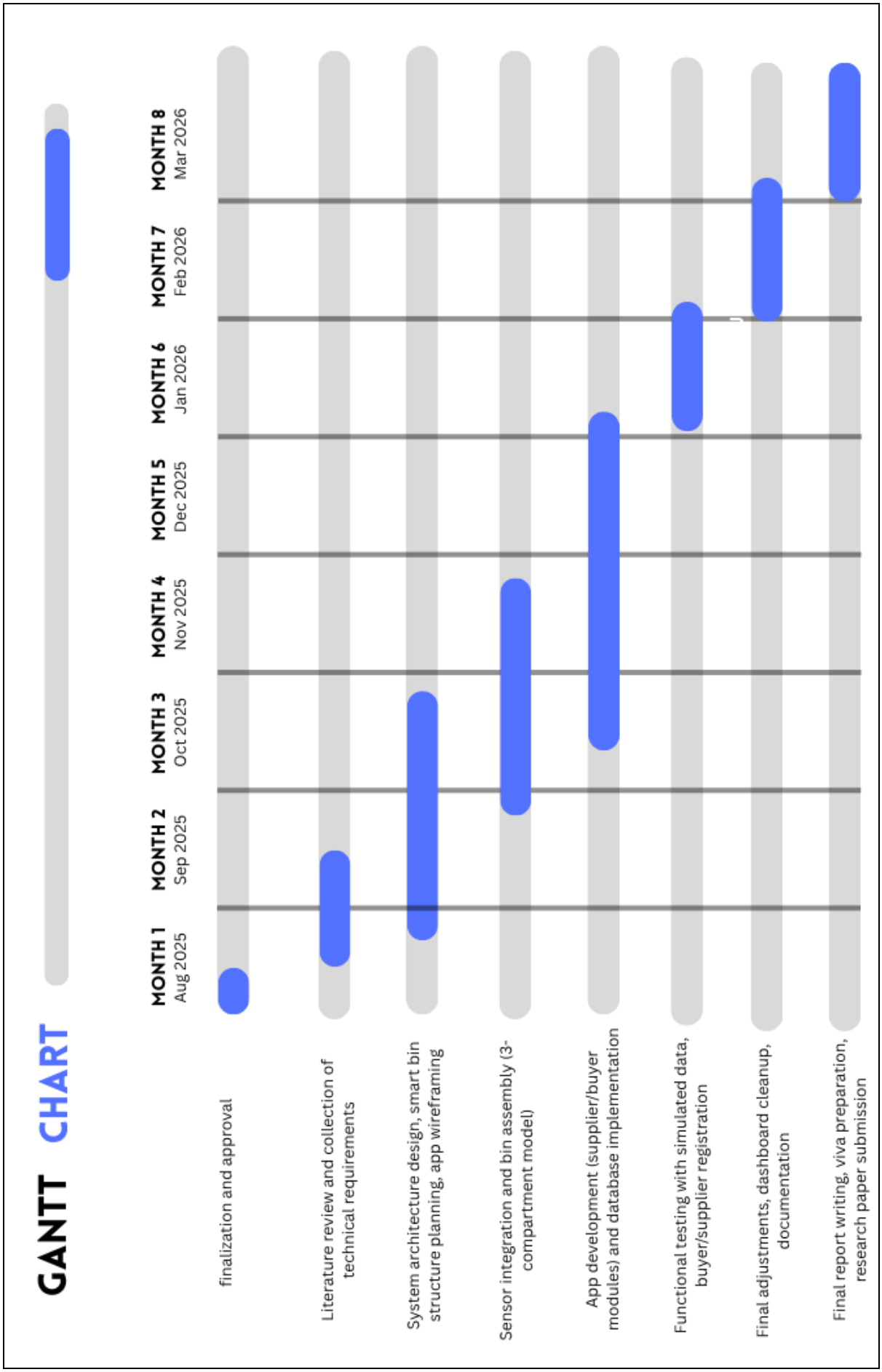
[5] A. Baloch and S. Zaidi, “Smart City Waste Management: IoT-Based Bin Monitoring System,” *ACM Transactions on Internet Technology (TOIT)*, vol. 21, no. 4, pp. 1–22, Aug. 2021. [Online]. Available: https://dl.acm.org/doi/abs/10.1145/3469887

[6] M. Ramesh and T. Krishnan, “Sensor fusion approaches in low-power smart bin applications,” *Journal of Cleaner Production*, vol. 276, p. 124122, 2020. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S0959652620345010

[7] H. S. Park, J. Kim, and M. Yoon, “Smart bin fill-level estimation using infrared and ultrasonic sensors,” *International Journal of Distributed Sensor Networks*, vol. 18, no. 6, pp. 1–12, June 2022. [Online]. Available: https://journals.sagepub.com/doi/10.1177/15501477221100263

[8] A. Shukla and N. Agarwal, “Design of IoT-Based Recyclable Waste Collection with Auction-Based Buyer Platform,” *IEEE Internet of Things Journal*, vol. 9, no. 5, pp. 3213–3222, Mar. 2022. [Online]. Available: https://ieeexplore.ieee.org/document/9631286

11. Gantt Chart


GANTT CHART
MONTH 1
Aug 2025
MONTH 2
Sep 2025
MONTH 3
Oct 2025
MONTH 4
Nov 2025
MONTH 5
Dec 2025
MONTH 6
Jan 2026
MONTH 7
Feb 2026
MONTH 8
Mar 2026
finalization and approval
Literature review and collection of technical requirements
System architecture design, smart bin structure planning, app wireframing
Sensor integration and bin assembly (3-compartment model)
App development (supplier/buyer modules) and database implementation
Functional testing with simulated data, buyer/supplier registration
Final adjustments, dashboard cleanup, documentation
Final report writing, viva preparation, research paper submission

Students Agreement:
By signing below on 2025/07/06, /We, as A.M.K.D ADHIKARI / W THILAN AVISHKA, (name/s) agree to the commonly known and accepted research ethics and guidelines. I/We commit to conducting the research project as mentioned above and outlined in the agreement. I/We understand that any changes to the group composition, identified research problem, or selected supervisor may result in penalties.

Name with Initials	Registration No	Signature
A.M.K.D ADHIKARI	ITBIN-2211-0137	Keshon
W. THILAN AVISHKA	ITBNM-2211-0199	

Supervisor(s) Agreement:
By signing below, I/We agree to supervise the aforementioned group of students throughout their research project, providing guidance and support for successful completion. Furthermore, by signing this document, I/we give my/our consent to the following: 1. The identified research problem is justifiable and feasible for investigation. 2. The research aim is achievable within a one-year timeframe. 3. The specified research objectives are measurable and achievable within the research period.

Supervisor's Name with Initials	Signature	Date
Ms. A.A.P.W Athukorala		2025/07/06
Mr. Daminda Herath		2025/07/06

Departmental Use Only:

Date Proposal Form Submitted: _____

All required sections of the form have been accurately completed, and the necessary signatures have been placed. The initial research proposal has been partially approved, thereby allowing the proposed research project to proceed in accordance with the specified guidelines outlined in the IT41028.

Research Coordinator's Signature:_____Date: ____

Department Head's Signature:_____Date: ____